

1)Write a Pandas program to select the distinct department id from

employees file.

Aim

To write a Python Pandas program to read the employees dataset and display the distinct (unique) Department IDs.

Procedure

1. Import the Pandas library.
2. Create or read the employees dataset.
3. Store the data in a Pandas DataFrame.
4. Select the **DEPARTMENT_ID** column.
5. Use the `unique()` function to obtain distinct department IDs.
6. Display the unique Department IDs.

Program

```
import pandas as pd
```

```
# Employee Department Data
```

```
data = {  
    "DEPARTMENT_ID": [10,20,30,40,50,60,70,80,90,100,110,  
                      120,130,140,150,160,170,180,190,200,  
                      210,220,230,240,250,260,270],  
    "DEPARTMENT_NAME": [  
        "Administration","Marketing","Purchasing","Human Resources",  
        "Shipping","IT","Public Relations","Sales","Executive",  
        "Finance","Accounting","Treasury","Corporate Tax",  
        "Control And Credit","Shareholder Services","Benefits",  
        "Manufacturing","Construction","Contracting","Operations",  
        "IT Support","NOC","IT Helpdesk","Government Sales",  
        "Retail Sales","Recruiting","Payroll"  
    ],  
    "MANAGER_ID": [200,201,114,203,121,103,204,145,100,108,205,  
                   0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0],  
    "LOCATION_ID": [1700,1800,1700,2400,1500,1400,2700,2500,  
                   1700,1700,1700,1700,1700,1700,1700,  
                   1700,1700,1700,1700,1700,1700,1700,  
                   1700,1700,1700,1700,1700]  
}
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Display unique Department IDs
```

```
print("Distinct Department IDs:")
```

```
print(df["DEPARTMENT_ID"].unique())
```

```
departments Table:
  DEPARTMENT_ID  DEPARTMENT_NAME  MANAGER_ID  LOCATION_ID
0             10      Administration         200           1700
1             20           Marketing         201           1800
2             30       Purchasing         114           1700
3             40   Human Resources         203           2400
4             50           Shipping         121           1500
5             60                IT         103           1400
6             70   Public Relations         204           2700
7             80            Sales         145           2500
8             90       Executive         100           1700
9            100            Finance         108           1700
10            110       Accounting         205           1700
11            120            Treasury          0           1700
12            130   Corporate Tax          0           1700
13            140  Control And Credit          0           1700
14            150  Shareholder Services          0           1700
15            160            Benefits          0           1700
16            170   Manufacturing          0           1700
17            180       Construction          0           1700
18            190       Contracting          0           1700
19            200       Operations          0           1700
20            210       IT Support          0           1700
21            220                NOC          0           1700
22            230       IT Helpdesk          0           1700
23            240  Government Sales          0           1700
24            250       Retail Sales          0           1700
25            260       Recruiting          0           1700
26            270           Payroll          0           1700

Distinct Department IDs:
0      10
1      20
2      30
3      40
4      50
5      60
6      70
7      80
8      90
9     100
10     110
11     120
12     130
13     140
```

2. Write a Pandas program to display the IDs of employees who have done two or more jobs in the past.

Aim

To write a Python Pandas program to identify employees who have worked in two or more jobs using the employee job history dataset.

Procedure

1. Import the Pandas library.
2. Create or read the employee job history dataset.
3. Store the data in a Pandas DataFrame.
4. Group the records by **EMPLOYEE_ID**.
5. Count the number of jobs for each employee.
6. Display employees whose job count is greater than or equal to 2.

Program

```
import pandas as pd

# Employee Job History Data
data = {
    "EMPLOYEE_ID": [102,101,101,201,114,122,200,176,176,200],
    "START_DATE": ["2001-01-13","1997-09-21","2001-10-28",
                  "2004-02-17","2006-03-24","2007-01-01",
                  "1995-09-17","2006-03-24","2007-01-01",
                  "2002-07-01"],
    "END_DATE": ["2006-07-24","2001-10-27","2005-03-15",
                "2007-12-19","2007-12-31","2007-12-31",
                "2001-06-17","2006-12-31","2007-12-31",
                "2006-12-31"],
    "JOB_ID": ["IT_PROG","AC_ACCOUNT","AC_MGR","MK_REP",
              "ST_CLERK","ST_CLERK","AD_ASST","SA_REP",
              "SA_MAN","AC_ACCOUNT"],
    "DEPARTMENT_ID": [60,110,110,20,50,50,90,80,80,90]
}

# Create DataFrame
df = pd.DataFrame(data)

# Count jobs done by each employee
job_count = df.groupby("EMPLOYEE_ID").size()
```

```
# Employees who have done two or more jobs
result = job_count[job_count >= 2]
```

```
print("Employees who have done two or more jobs:")
print(result)
```

```
-----RESTART: C:/Users/subha/employees.py-----
Job History Table:
```

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
0	102	2001-01-13	2006-07-24	IT_PROG	60
1	101	1997-09-21	2001-10-27	AC_ACCOUNT	110
2	101	2001-10-28	2005-03-15	AC_MGR	110
3	201	2004-02-17	2007-12-19	MK_REP	20
4	114	2006-03-24	2007-12-31	ST_CLERK	50
5	122	2007-01-01	2007-12-31	ST_CLERK	50
6	200	1995-09-17	2001-06-17	AD_ASST	90
7	176	2006-03-24	2006-12-31	SA_REP	80
8	176	2007-01-01	2007-12-31	SA_MAN	80
9	200	2002-07-01	2006-12-31	AC_ACCOUNT	90

Employee IDs who have done two or more jobs:

```
[101, 200, 176]
```

3) Write a Pandas program to display the details of jobs in descending sequence on job title.

Aim

To write a Python Pandas program to display the details of jobs in descending order based on the JOB_TITLE.

Procedure

1. Import the Pandas library.
2. Create or read the jobs dataset.
3. Store the data in a Pandas DataFrame.
4. Sort the DataFrame by the JOB_TITLE column in descending order using `sort_values()`.
5. Display the sorted DataFrame.

Program

```
import pandas as pd
```

```
# Job Data
```

```
data = {
```

```
    "JOB_ID": [
```

```
        "AD_PRES","AD_VP","AD_ASST","FI_MGR","FI_ACCOUNT",
```

```
        "AC_MGR","AC_ACCOUNT","SA_MAN","SA_REP","PU_MAN",
```

```
        "PU_CLERK","ST_MAN","ST_CLERK","SH_CLERK","IT_PROG",
```

```
        "MK_MAN","MK_REP","HR_REP","PR_REP"
```

```
    ],
```

```
    "JOB_TITLE": [
```

```
        "President","Administration Vice President",
```

```
        "Administration Assistant","Finance Manager",
```

```
        "Accountant","Accounting Manager",
```

```
        "Public Accountant","Sales Manager",
```

```
        "Sales Representative","Purchasing Manager",
```

```
        "Purchasing Clerk","Stock Manager",
```

```
        "Stock Clerk","Shipping Clerk",
```

```
        "Programmer","Marketing Manager",
```

```
        "Marketing Representative",
```

```
        "Human Resources Representative",
```

```
        "Public Relations Representative"
```

```
    ],
```

```
    "MIN_SALARY": [
```

```

20080,15000,3000,8200,4200,8200,4200,
10000,6000,8000,2500,5500,2008,2500,
4000,9000,4000,4000,4500
],

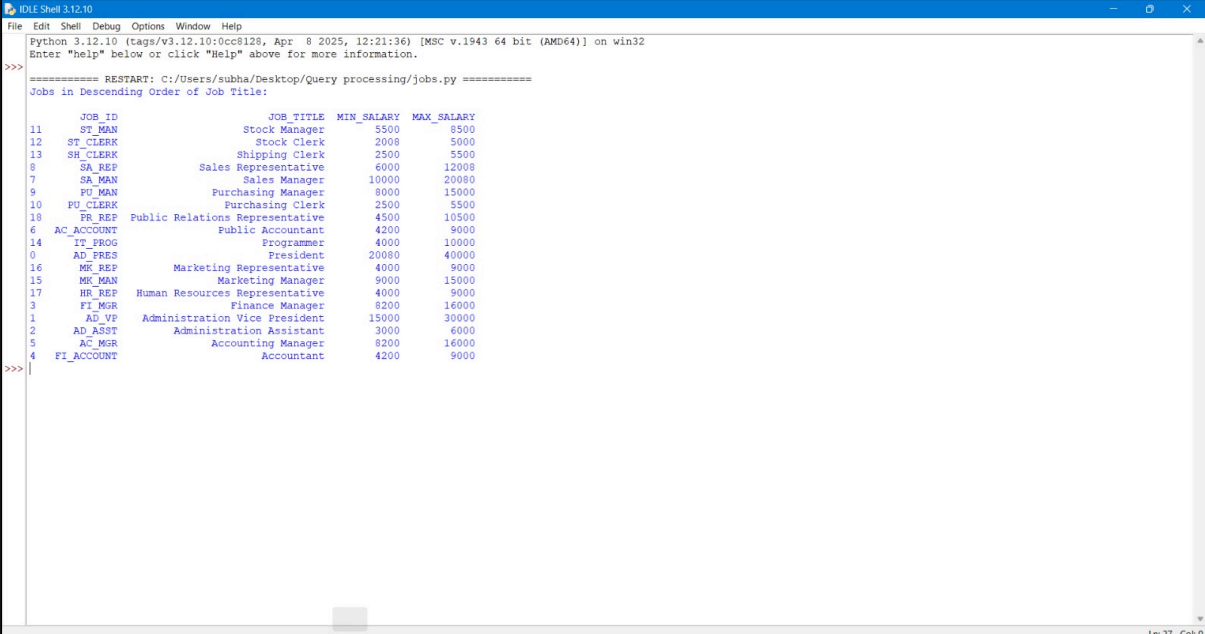
"MAX_SALARY": [
    40000,30000,6000,16000,9000,16000,9000,
    20080,12008,15000,5500,8500,5000,5500,
    10000,15000,9000,9000,10500
]
}

# Create DataFrame
df = pd.DataFrame(data)

# Sort by JOB_TITLE in descending order
result = df.sort_values(by="JOB_TITLE", ascending=False)

print(result)

```



```

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/subha/Desktop/Query processing/jobs.py =====
Jobs in Descending Order of Job Title:

```

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
11	ST_MAN Stock Manager	5500	8500
12	ST_CLERK Stock Clerk	2008	5000
13	SH_CLERK Shipping Clerk	2500	5500
8	SA_REP Sales Representative	6000	12008
7	SA_MAN Sales Manager	10000	20080
9	PU_MAN Purchasing Manager	8000	15000
10	PU_CLERK Purchasing Clerk	2500	5500
18	PR_REP Public Relations Representative	4500	10500
6	AC_ACCOUNT Public Accountant	4200	9000
14	IT_PROG Programmer	4000	10000
0	AD_PRES President	20080	40000
16	MK_REP Marketing Representative	4000	9000
15	MK_MAN Marketing Manager	9000	15000
17	HR_REP Human Resources Representative	4000	9000
3	FI_MGR Finance Manager	8200	16000
1	AD_VP Administration Vice President	15000	30000
2	AD_ASST Administration Assistant	3000	6000
5	AC_MGR Accounting Manager	8200	16000
4	FI_ACCOUNT Accountant	4200	9000